Mastering Generative AI & LLM Engineering

From Foundations to Applications

Led by : Shrevas Shukla

in association with

iHUB DivyaSampark, IIT Roorkee & RBPL

Mastering Generative AI & LLM Engineering

From Foundations to Applications

✓ AI Concepts

✓ Python Programming

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- ✓ AI Concepts
- ✓ Python Programming
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Mastering Generative AI & LLM Engineering
From Foundations to Applications

- ✓ AI Concepts
- ✓ Python Programming
- ✓ Natural Language Programming
- ✓ Large Language Model

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Shreyas Shukla

Freelance AI Trainer and Consultant

- Ritvij Bharat Private Limited (RBPL)
- iHUB DivyaSampark, IIT Roorkee
- E & ICT Academy, IIT Guwahati
- Daksh Gurukul, IIT Guwahati
- MASAI , Bangalore
- Edubridge Learning Pvt. Ltd.
- Indium Software

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https://tih.iitr.ac.in/training-courseDetails/103

Conducted by Dr. Chaitanya Shukla

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Mastering Generative AI & LLM Engineering
From Foundations to Applications
Lecture by Shreyas Shukla
in association with
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Please don’t skip sessions

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Evolution of AI

Led by Dr. Shreyas Shukla

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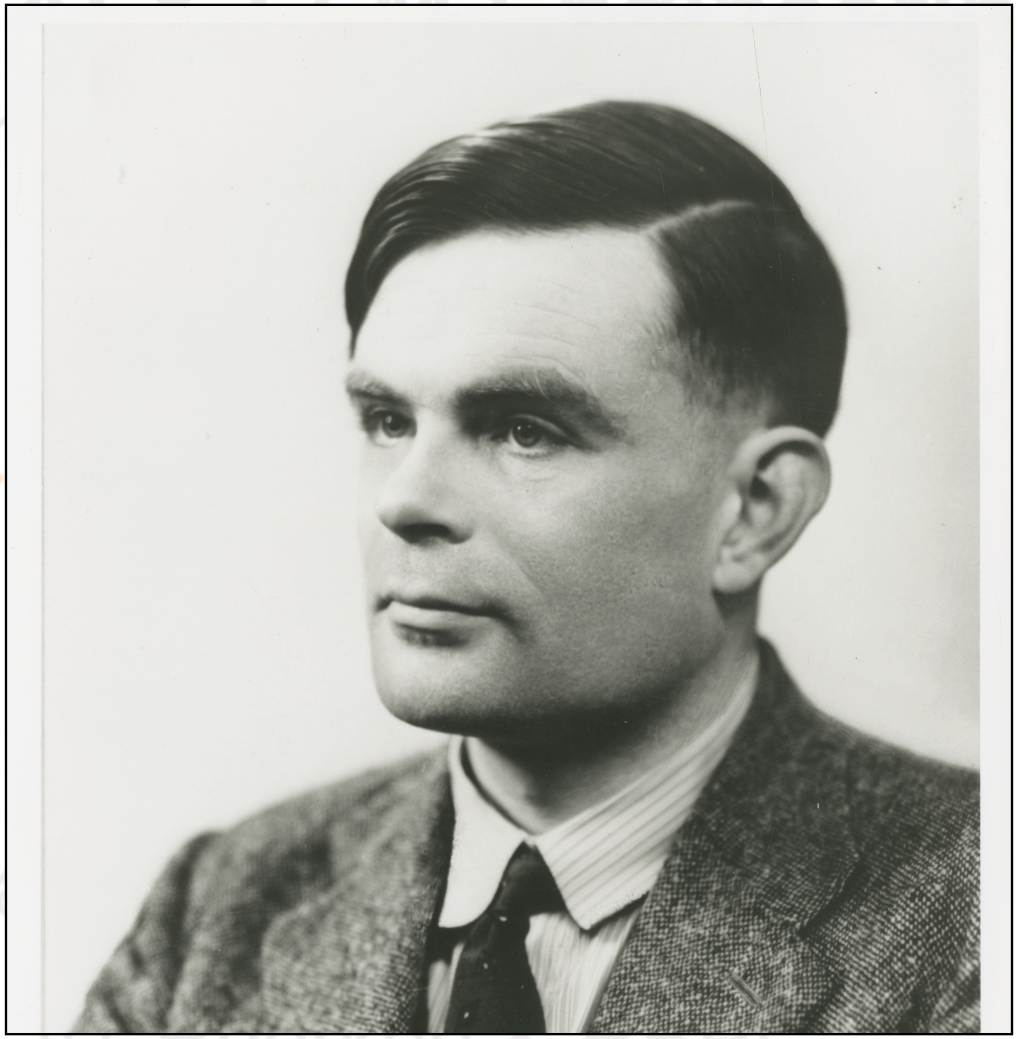
1950s : Can Machines think ?

Alan Turing

(1912 - 1954)



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The Turing Test

A human evaluator communicates in natural language with both a human and a machine, without seeing either of them

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If the evaluator cannot reliably tell which participant is the machine, the machine is considered to have passed the test, showing human-like intelligence.

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1956 : Term ‘AI’ was coined at Dartmouth Conference

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1997 : IBM's Deep Blue defeated world chess champion Garry Kasparov

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The Times

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Deep Blue Defeats Kasparaiv!

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Geoffrey Hinton
‘Godfather of AI’

Published a paper in 2006 on Neural Networks

2012 : Deep Neural Networks

Appendix: Building high-level features using large scale unsupervised learning

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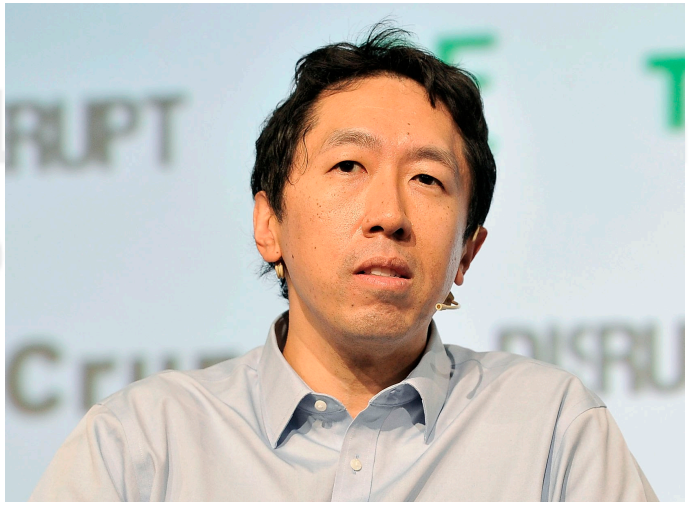
Jeff Dean JEFF@GOOGLE.COM
Andrew Y. Ng ANG@CS.STANFORD.EDU



Andrew Ng

Jeff Dean

Abstract

In this appendix, we discuss more details regarding the algorithm, its implementation, test set for 3D-transformed faces, experimental results for parameter sensitivity. We also present further visualizations for the learned neurons.

A. Training and test images

A subset of training images is shown in Figure 1. As can be seen, the positions, scales, orientations of faces in the dataset are diverse. A subset of test images for identifying the face neuron is shown in Figure 2.

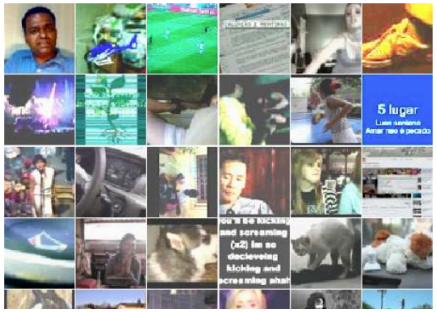


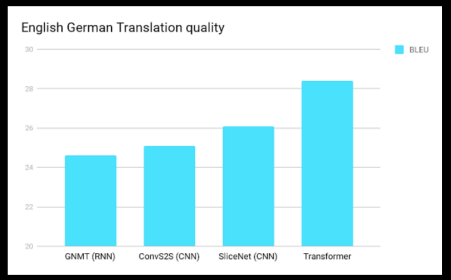
Figure 2. Some example test set images (shown before the whitening step).

B. Models

Central to our approach in this paper is the use of locally-connected networks. In these networks, neurons only connect to a local region of the layer below.

In Figure 3, we show the connectivity patterns of the neural network architecture described in the paper.

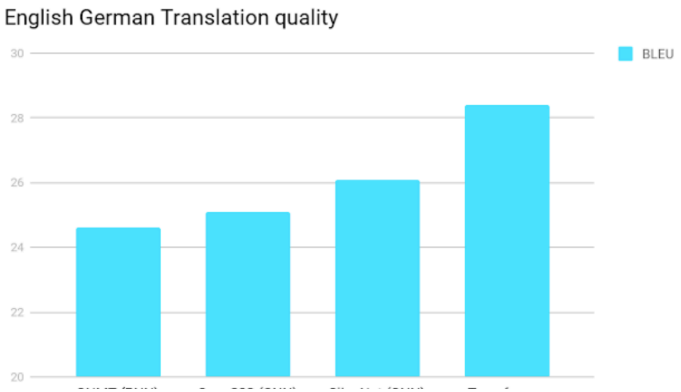
Transformer: A Novel Neural Network Architecture for Language Understanding



August 31, 2017 · Posted by Jakob Uszkoreit, Software Engineer, Natural Language Understanding

Neural networks, in particular recurrent neural networks (RNNs), are now at the core of the leading approaches to language understanding tasks such as language modeling, machine translation and question answering. In "Attention Is All You Need", we introduce the Transformer, a novel neural network architecture based on a self-attention mechanism that we believe to be particularly well suited for language understanding.

In our paper, we show that the Transformer outperforms both recurrent and convolutional models on academic English to German and English to French translation benchmarks. On top of higher translation quality, the Transformer requires less computation to train and is a much better fit for modern machine learning hardware, speeding up training by up to an order of magnitude.



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2018

OpenAI released the first version of GPT series

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2025

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OpenAI introduces GPT-5

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AI vs ML vs DL

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vs DS vs DA vs DE vs CV vs NLP..... etc. etc.

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Artificial Intelligence

AI involves techniques that equip computers to emulate human behavior, enabling them to learn, make decisions, recognize patterns, and solve complex problems in a manner akin to human intelligence.

Machine Learning

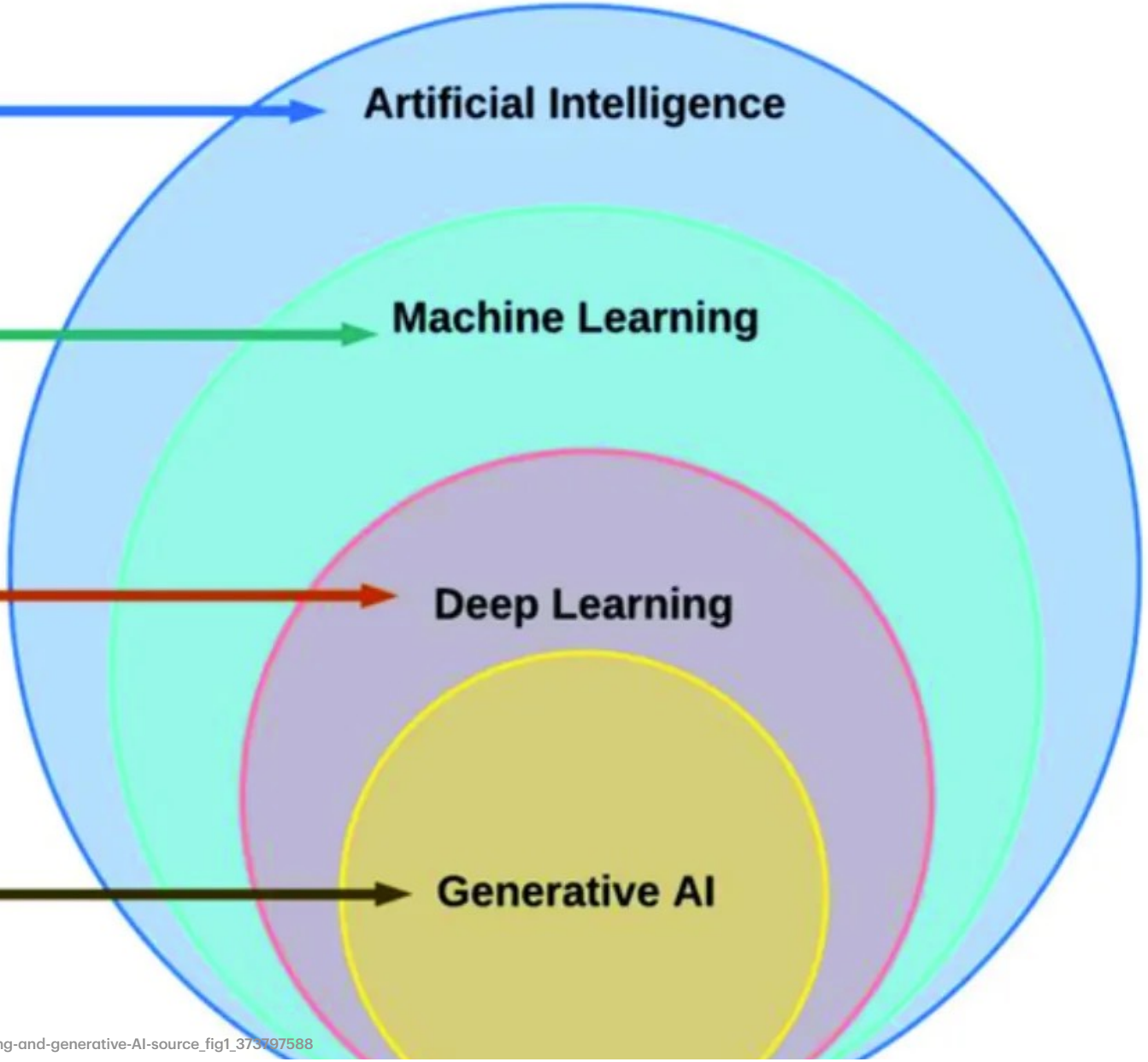
ML is a subset of AI, uses advanced algorithms to detect patterns in large data sets, allowing machines to learn and adapt. ML algorithms use supervised or unsupervised learning methods.

Deep Learning

DL is a subset of ML which uses neural networks for in-depth data processing and analytical tasks. DL leverages multiple layers of artificial neural networks to extract high-level features from raw input data, simulating the way human brains perceive and understand the world.

Generative AI

Generative AI is a subset of DL models that generates content like text, images, or code based on provided input. Trained on vast data sets, these models detect patterns and create outputs without explicit instruction, using a mix of supervised and unsupervised learning.



Source : https://www.researchgate.net/figure/A-comparative-view-of-AI-machine-learning-deep-learning-and-generative-AI-source_fig1_373797588

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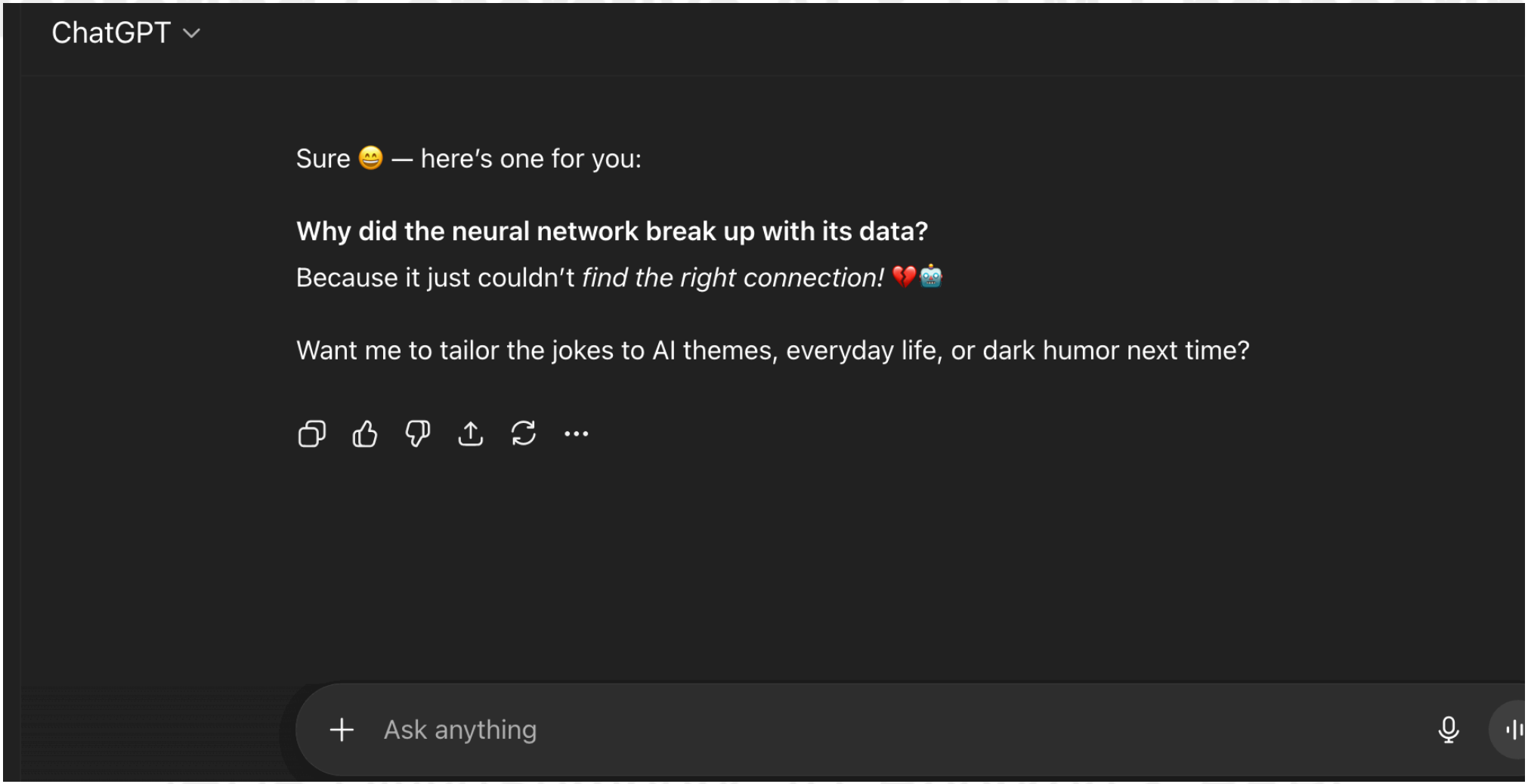
Weak vs Strong AI


ChatGPT
Sure 😄 — here's one for you:
Why did the neural network break up with its data?
Because it just couldn't find the right connection! 💔🤖
Want me to tailor the jokes to AI themes, everyday life, or dark humor next time?
Ask anything

Sam Altman

CEO, OpenAI

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Machine Learning

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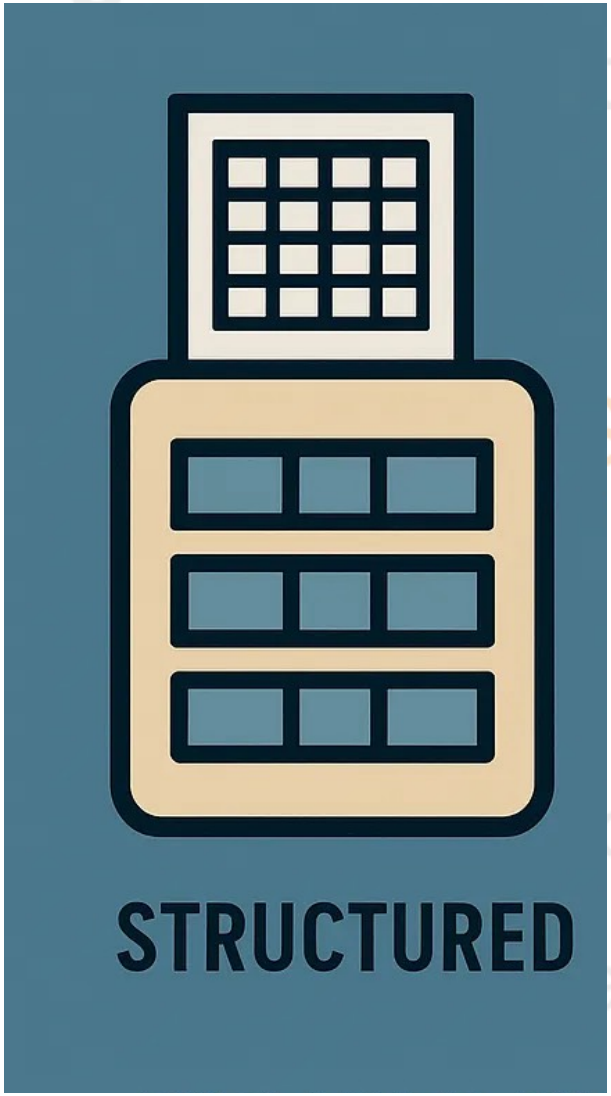
STRUCTURED VS UNSTRUCTURED DATA



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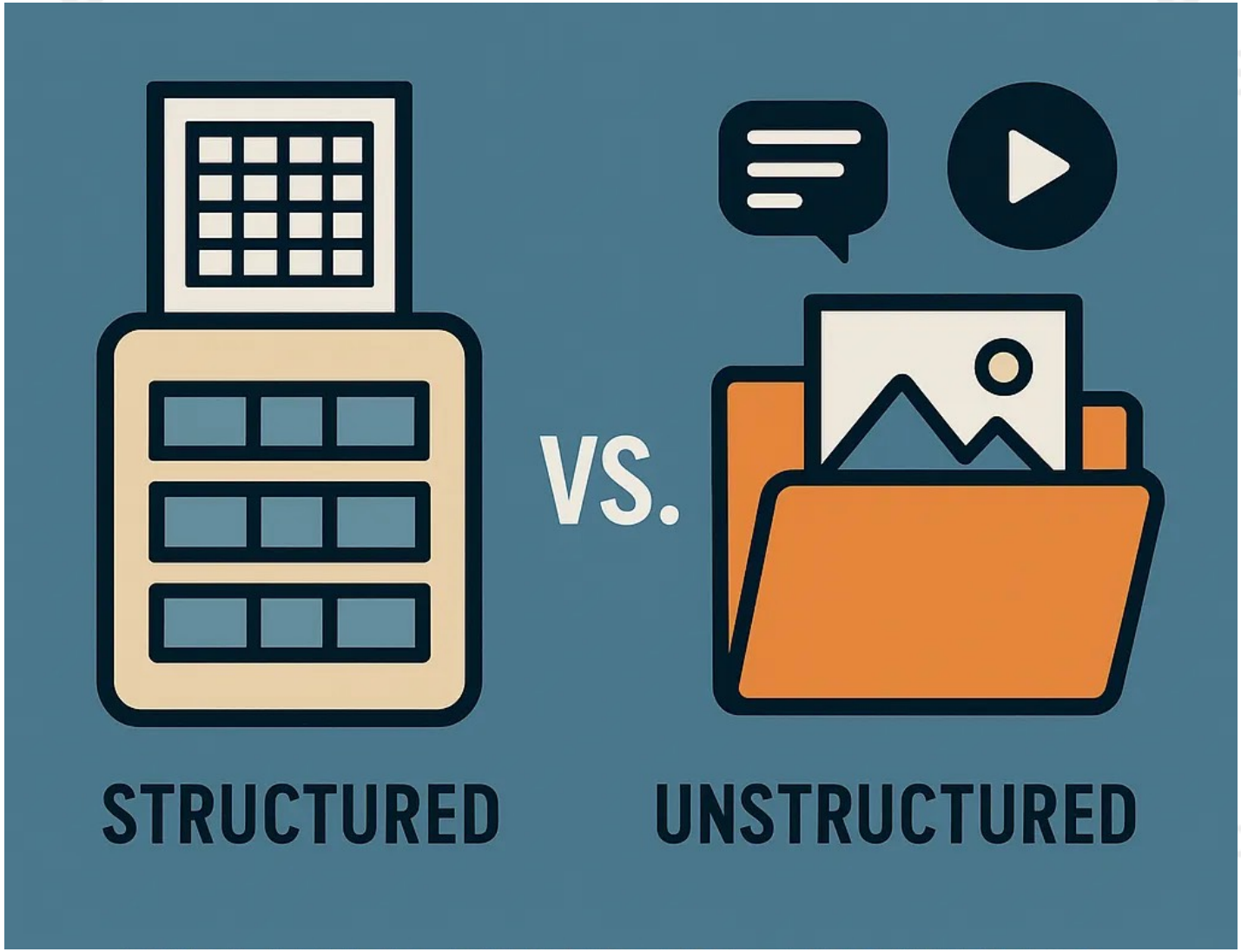
STRUCTURED VS UNSTRUCTURED DATA



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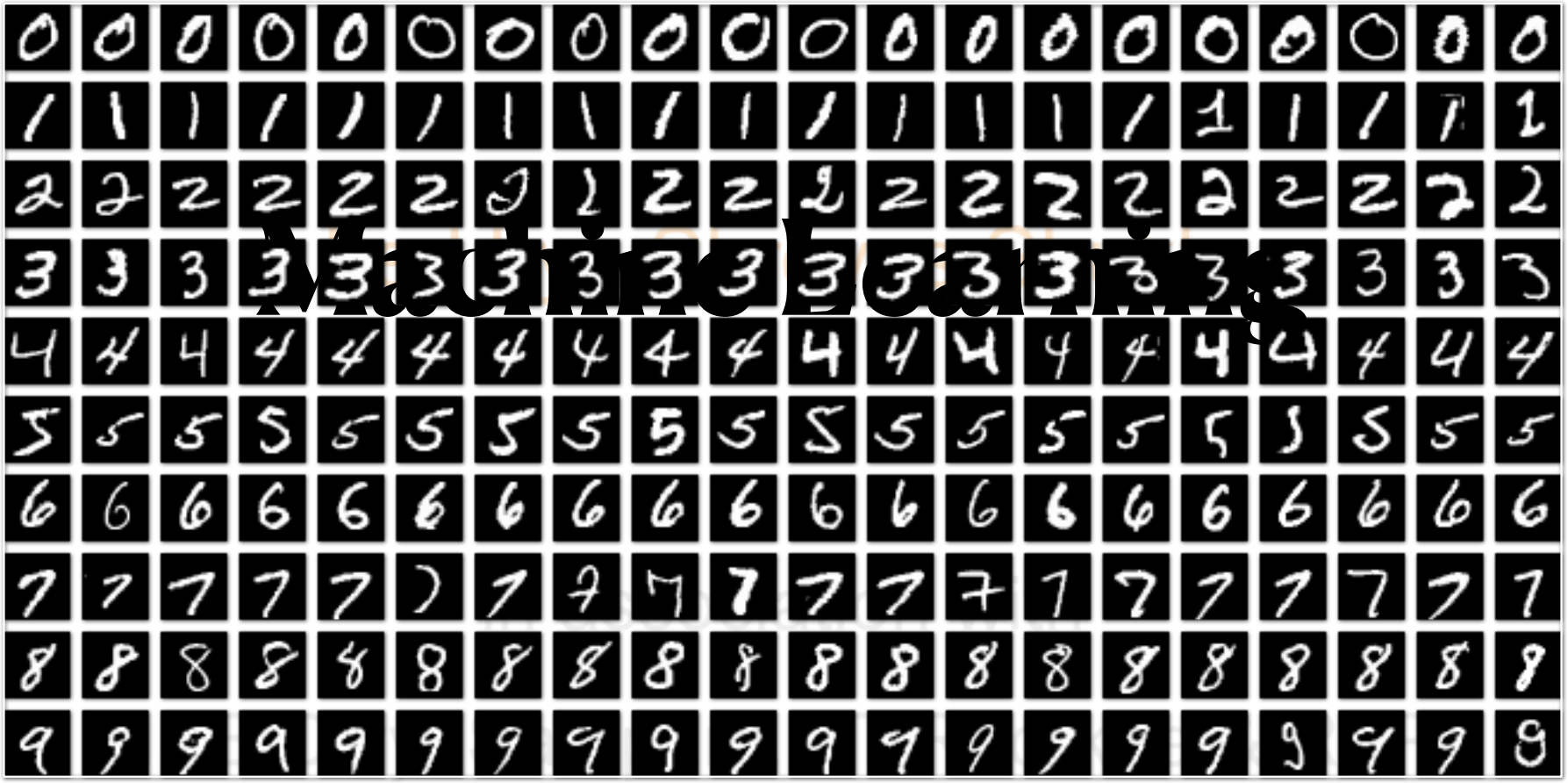
MNIST database

70000 28*28 images of handwritten digits



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Machine Learning

1. Videos are made of thousands of images, each containing many pixels.
2. Each pixel represents a number stored in binary (zeros and ones).
3. Sound and written text are also represented as sequences of zeros and ones.
4. Binary format gives computers the structure needed for learning.
5. Enables pattern recognition, similarity detection, and task execution.

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LABELLED VS UNLABELLED DATA

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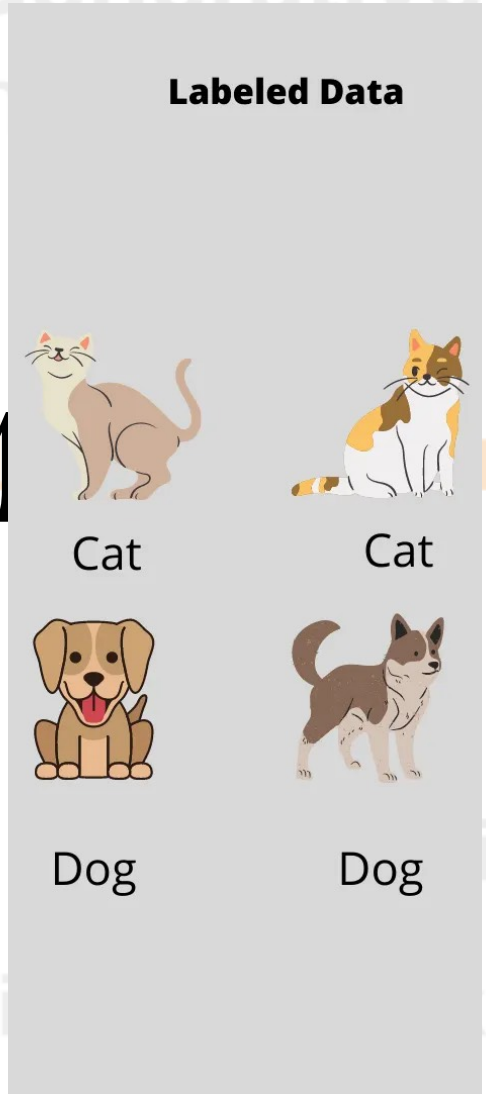
LABELLED VS UNLABELLED DATA



M Learning

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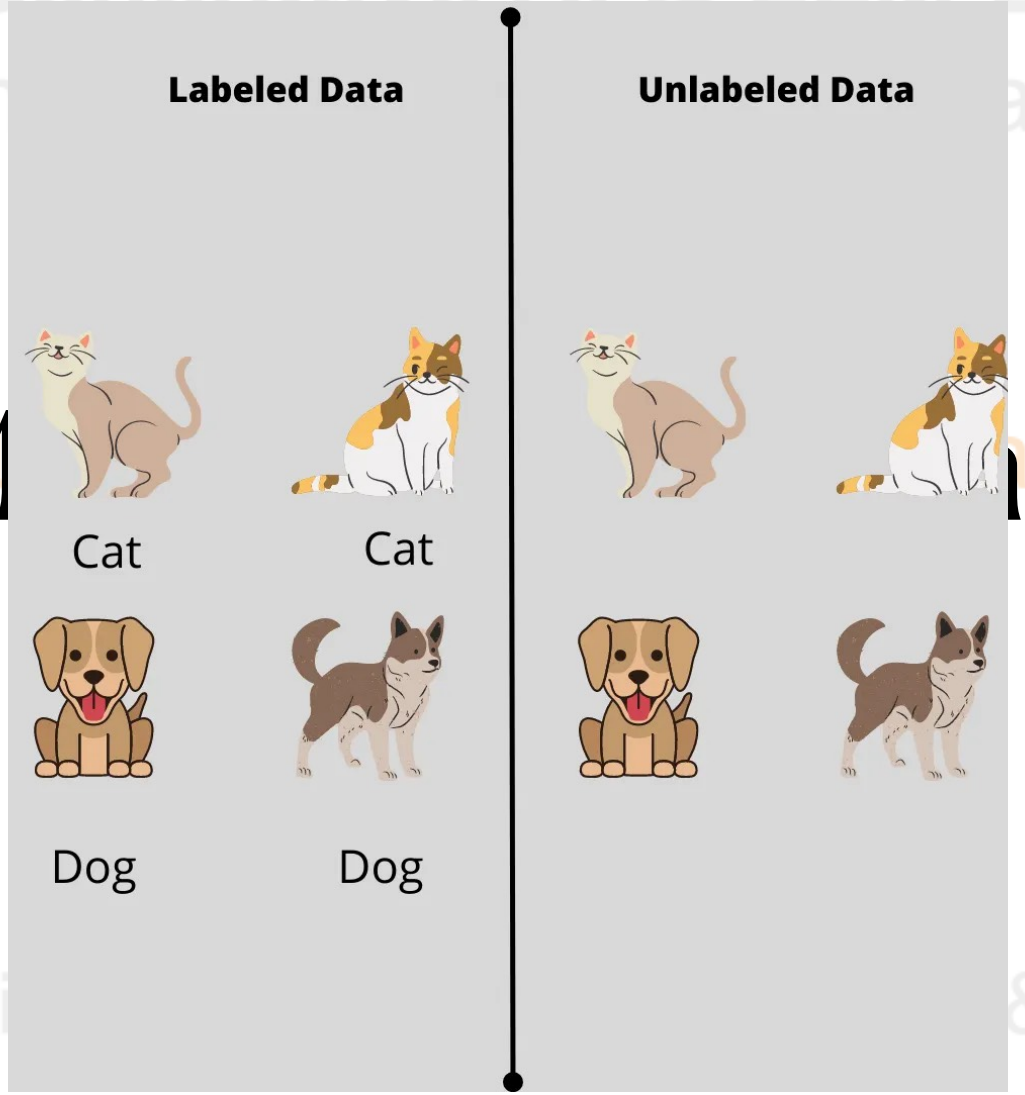
LABELLED VS UNLABELLED DATA



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- Machine learning enables systems to learn and improve through trial and error.
- Quality training data is essential for effective learning.
- Goal: prepare models to solve problems with unseen data.
- More good data can outweigh model complexity in performance.
- Updating algorithms and training data enhances accuracy and adaptability.
- Continuous learning is vital for robust, versatile AI systems.
- Different ML models are suited for specific problem domains.

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- Real estate agent wants to develop an app to estimate home selling prices.
- Data scientist checks if historical sales data is available.
- ML model trained on historical data to find patterns.
- Input (*x*): home characteristics; Output (*y*): predicted price.
- Model predicts property value for new user inputs using learned patterns.

X			y
Area m^2	Bedrooms	Bathrooms	Price
200	2	2	Rs.50,00,000
190	1	1	Rs. 40, 50,000
230	3	2	Rs. 60, 50,000
180	2	1	Rs. 40, 00,000
210	3	1	Rs. 50, 50,000

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Supervised, Unsupervised and Reinforcement Learning

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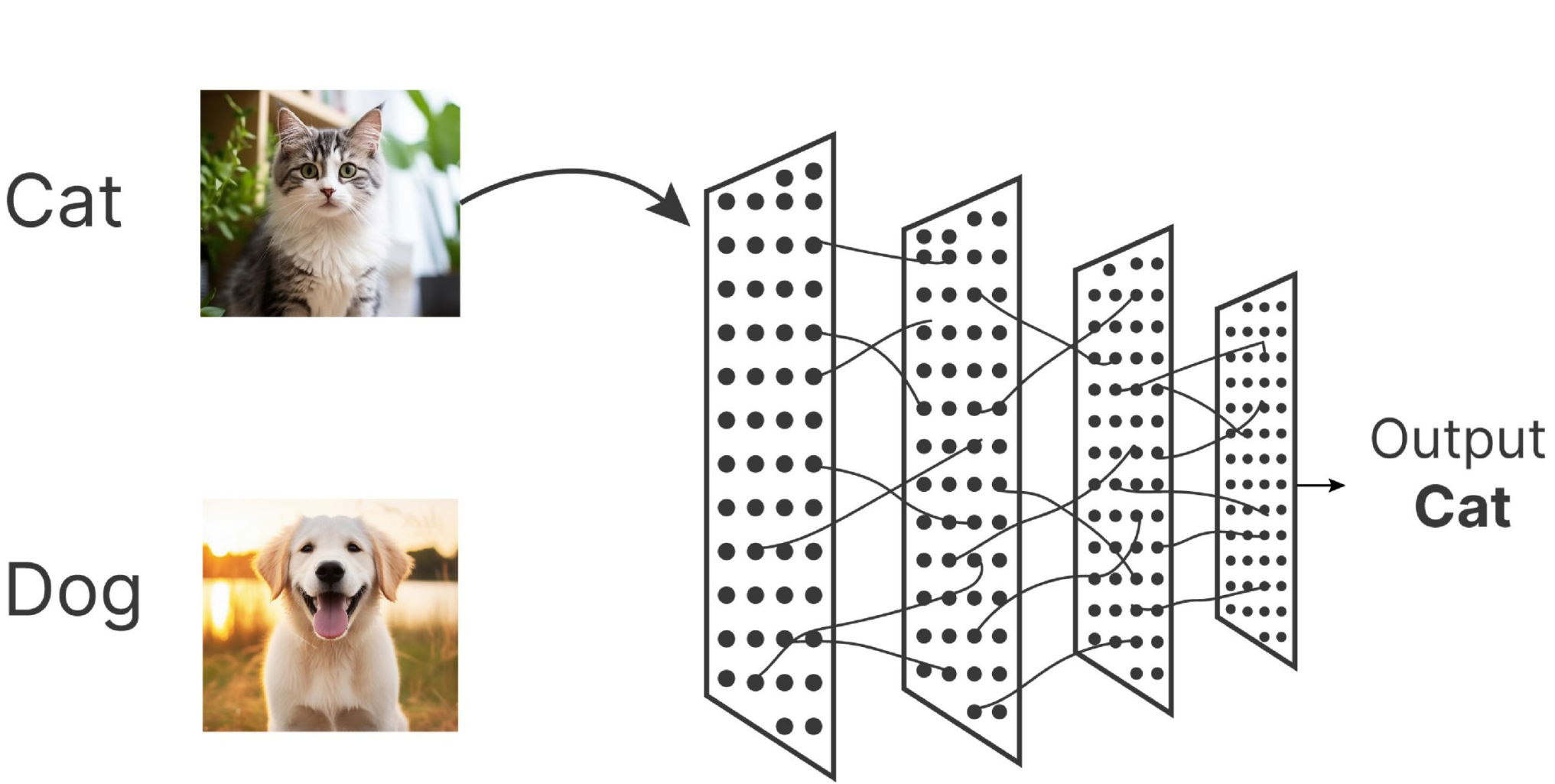
Supervised Learning (Classification)



Supervised Learning (Regression)

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	X		y
Area m^2	Bedrooms	Bathrooms	Price
200	2	2	Rs.50,00,000
190	1	1	Rs. 40, 50,000
230	3	2	Rs. 60, 50,000
180	2	1	Rs. 40, 00,000
210	3	1	Rs. 50, 50,000

Mastering Generative AI & LLM Engineering
From Foundations to Applications

Unsupervised Learning in ML

In association with
iHUB DivyaSampark, IIT Roorkee & RBPL

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From Foundations to Applications
Lead by: Dr. Shivam Shukla
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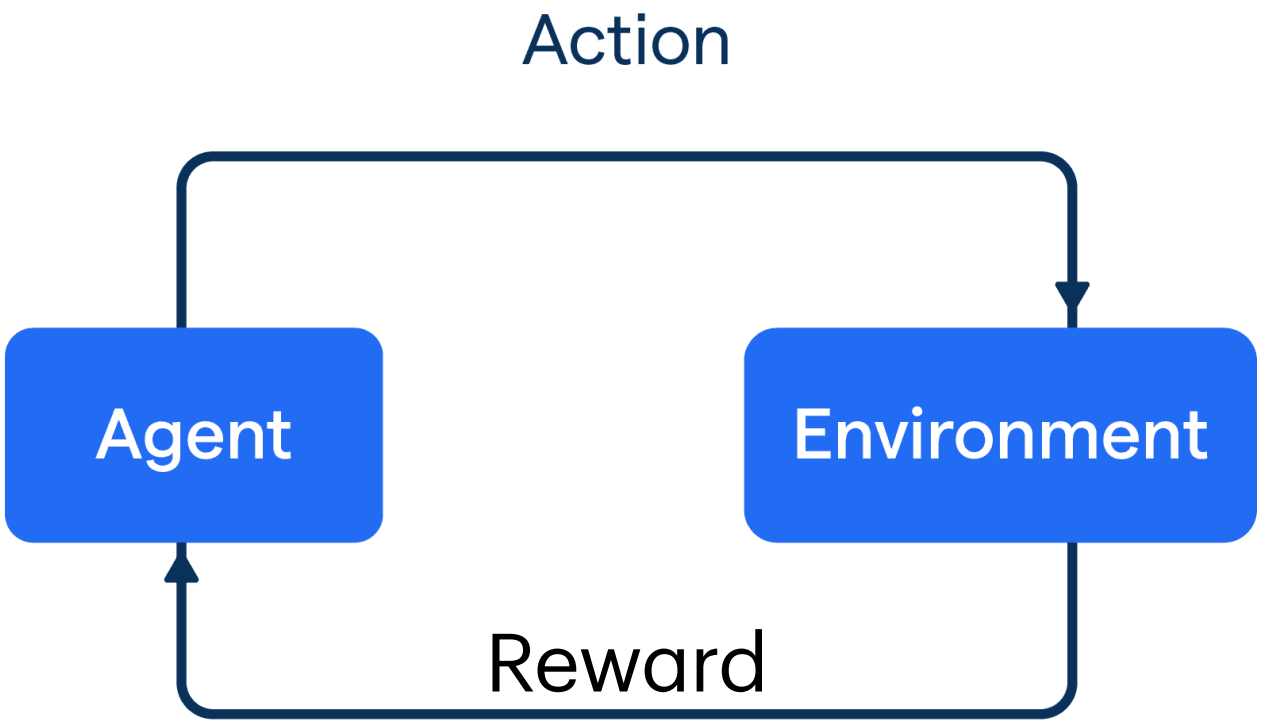
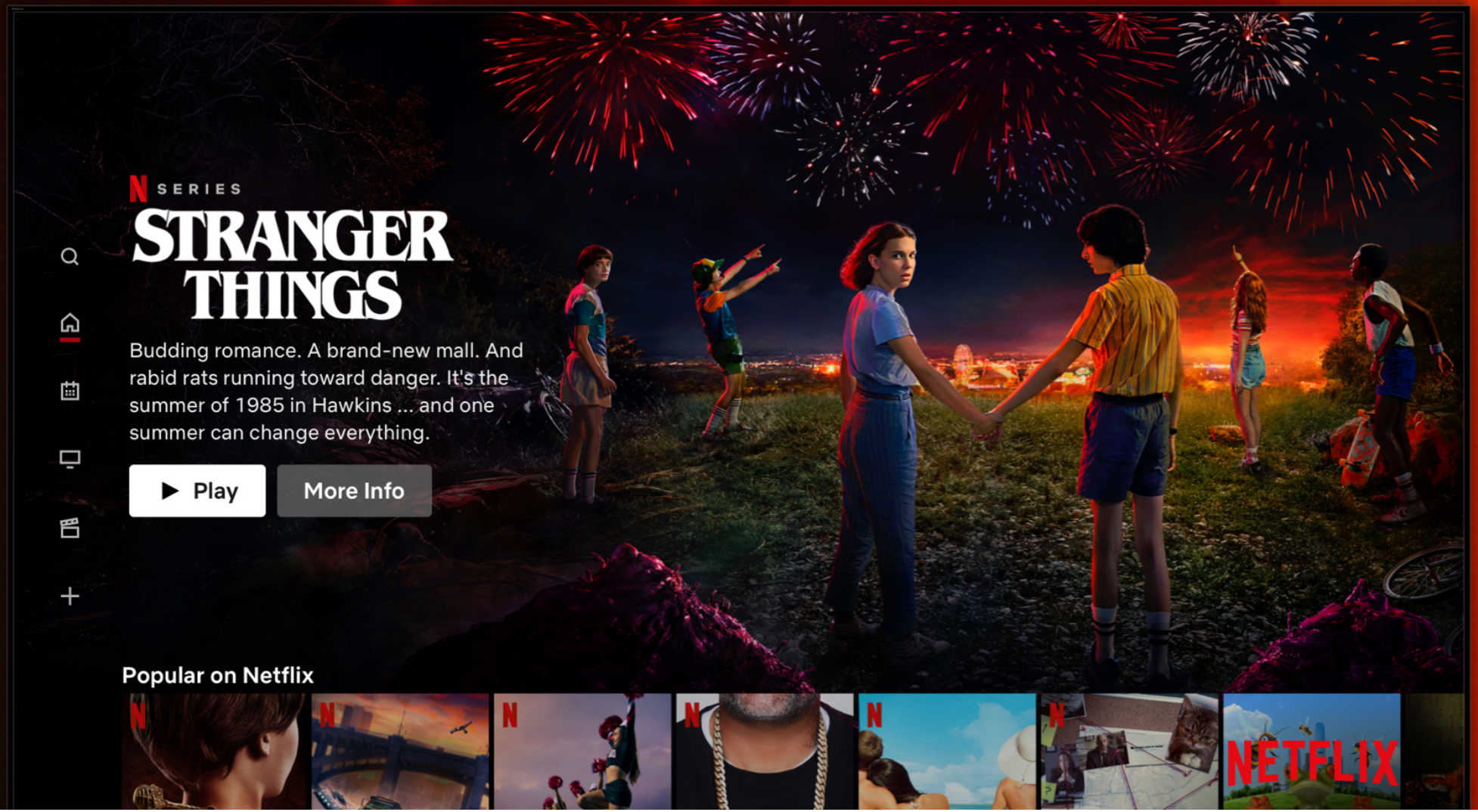
Labelling Data is expensive

Reinforcement Learning

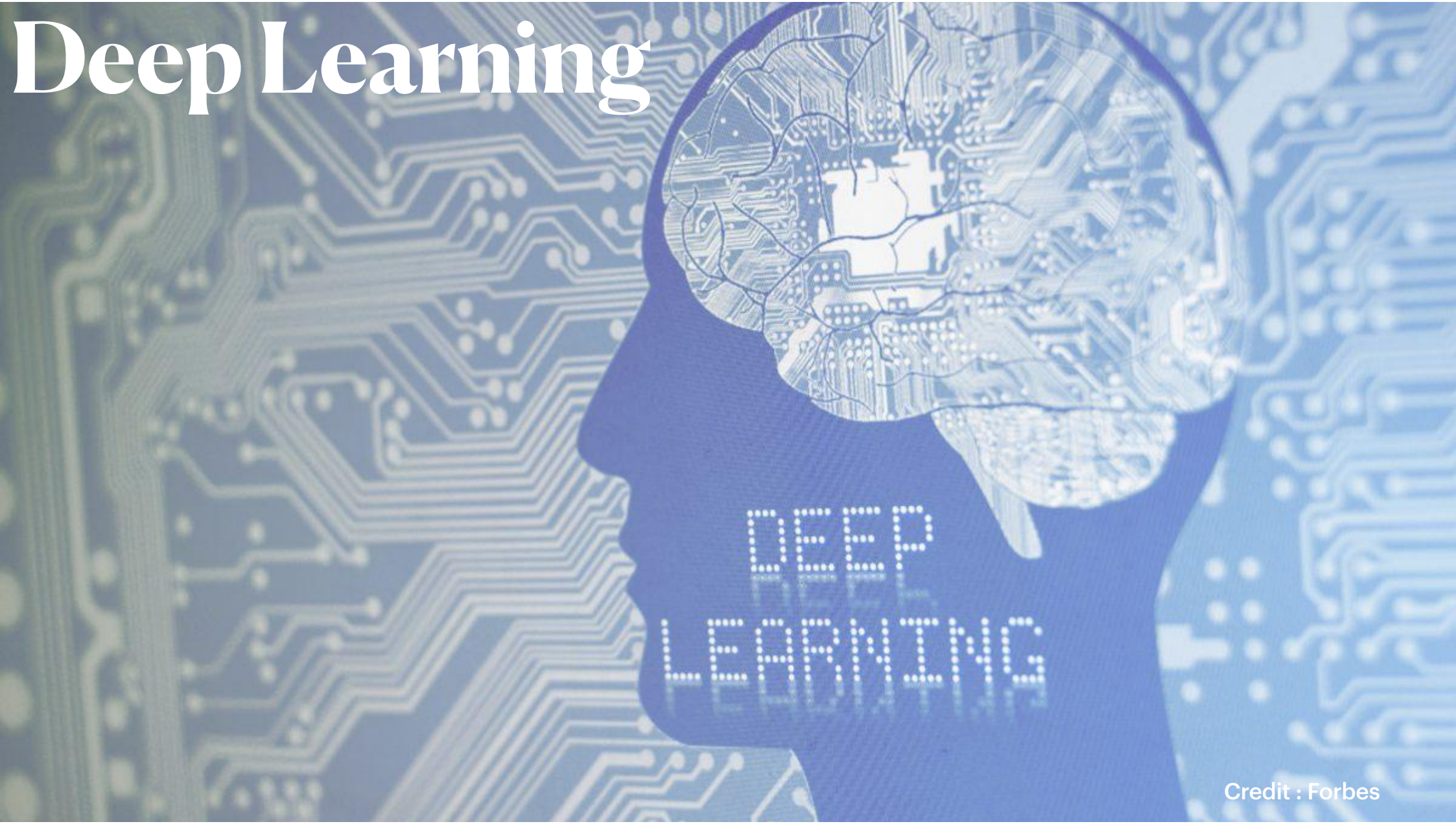


Unlimited Entertainment

Watch all you want for one low price.



Deep Learning



Credit : Forbes

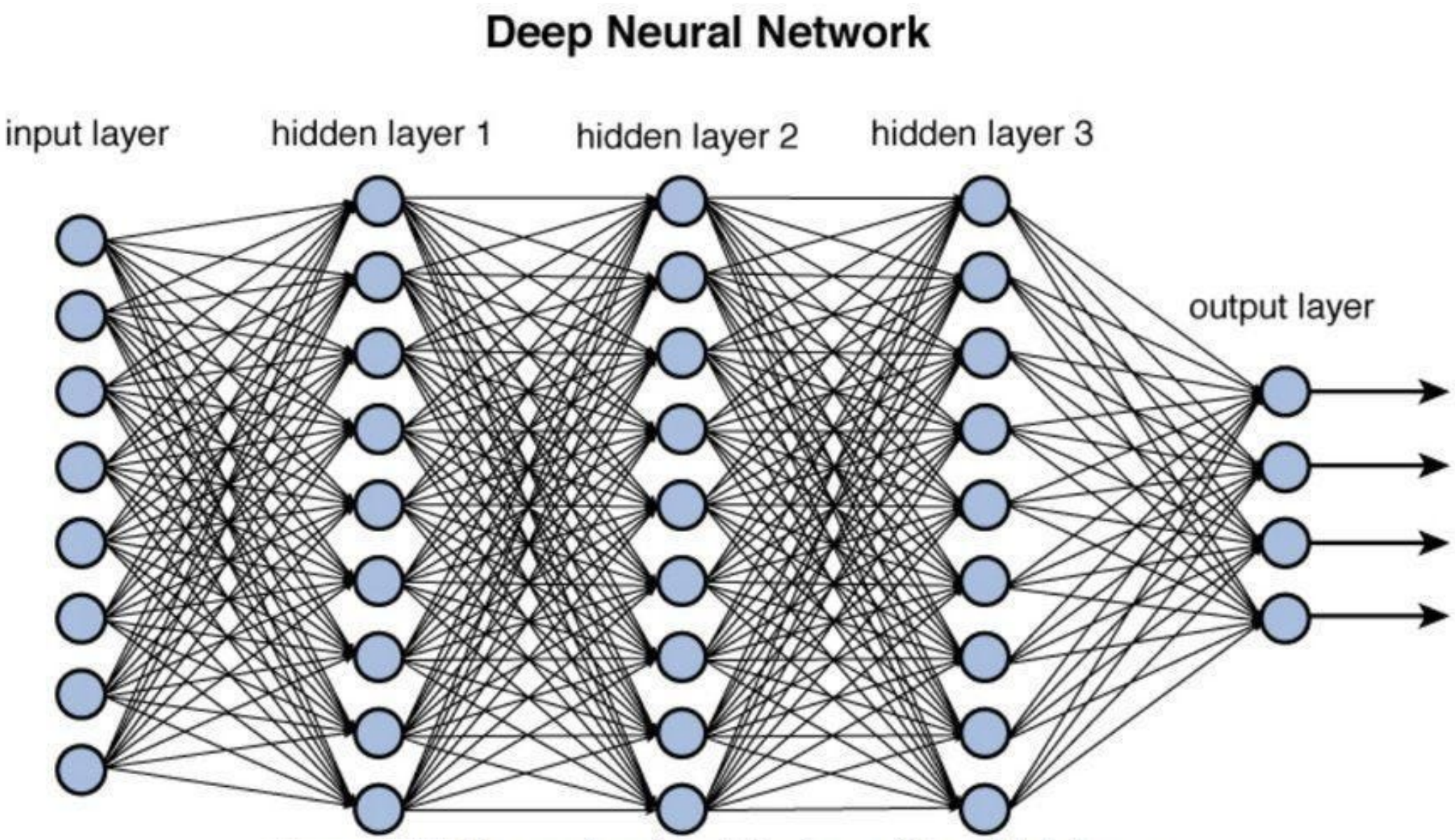
ChatGPT

Share

Generate an image of an crowded but clean beach in India

Ask anything

ChatGPT can make mistakes. Check important info. See Cookie Preferences.



Deep network architecture with multiple layers.

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Led by: Shreyas Shukla
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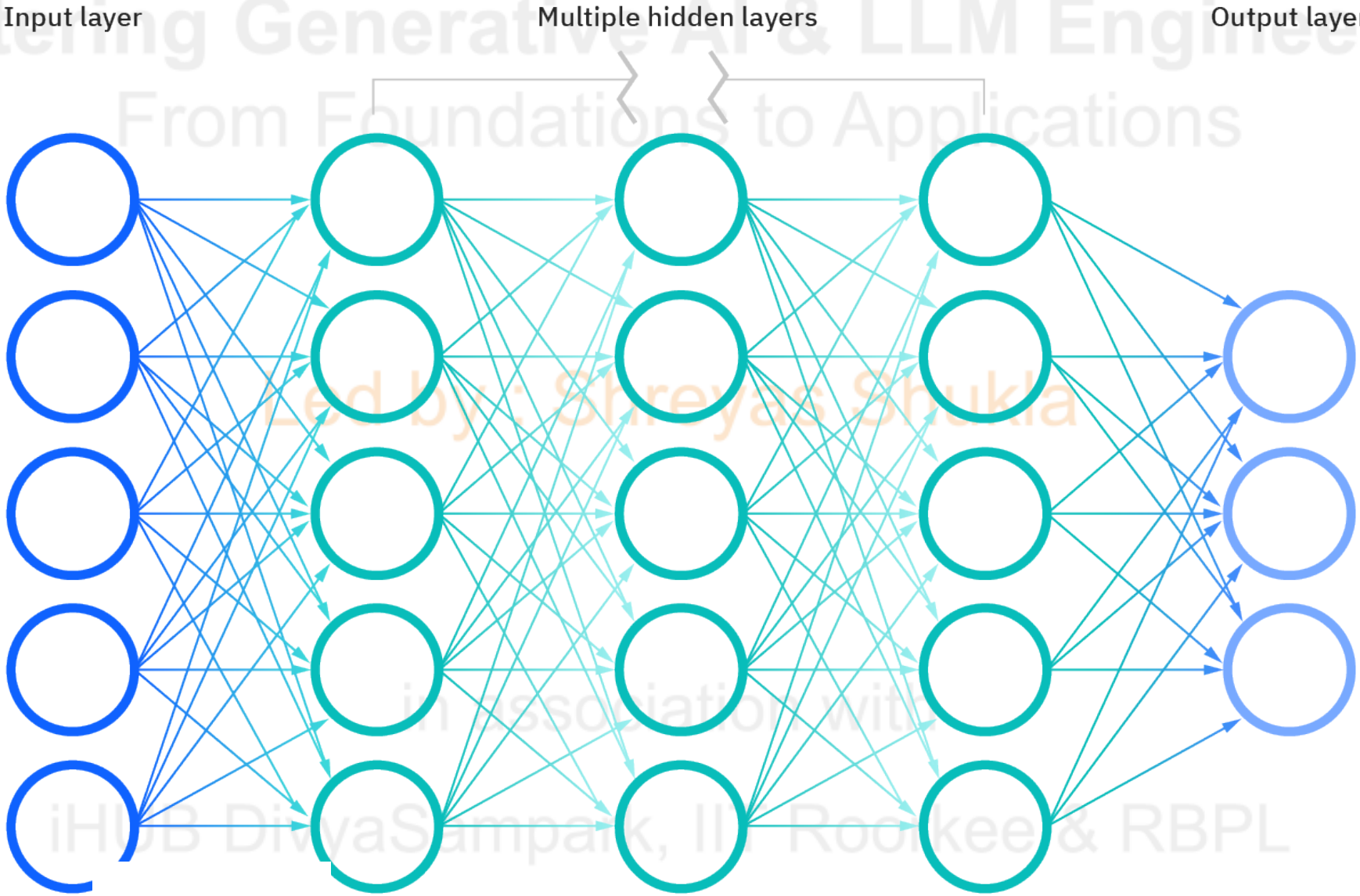
Deep neural network



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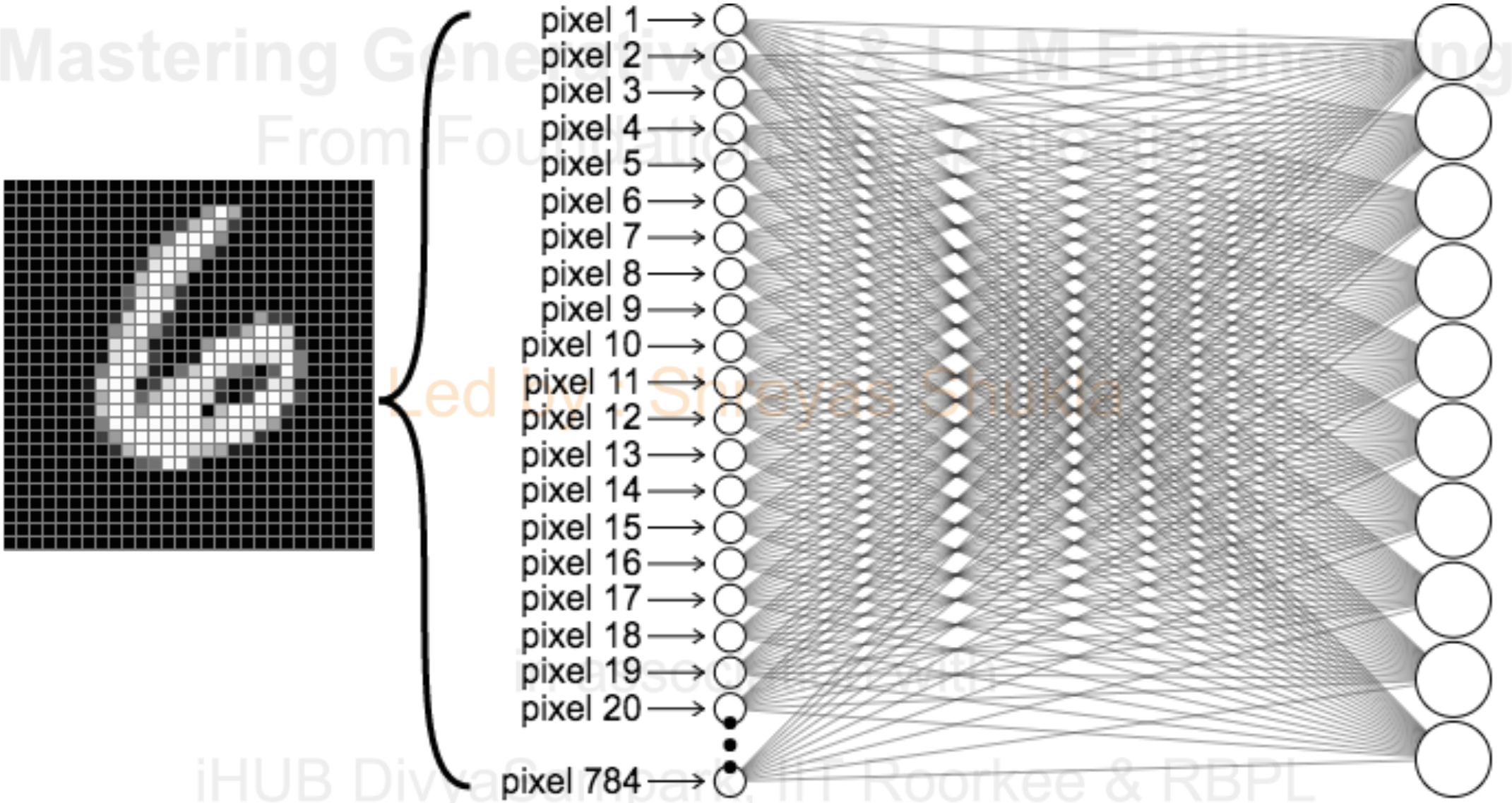
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